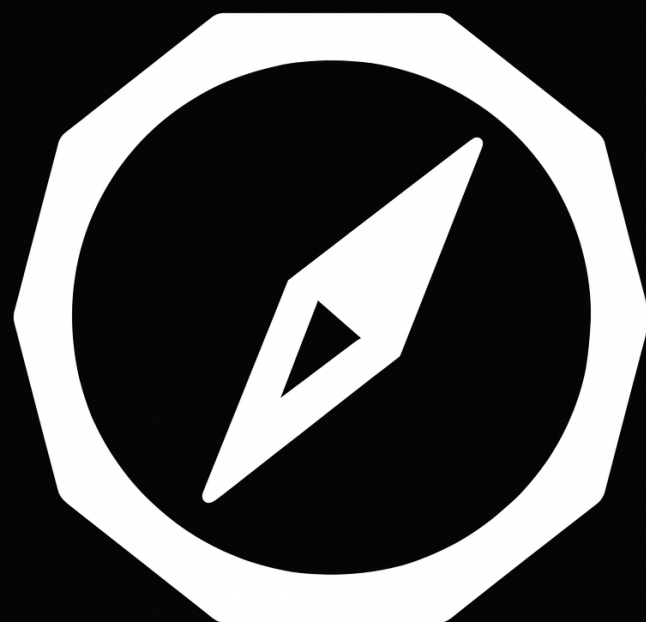


WEAK7 TEAM4



김윤수
김희준
홍신화

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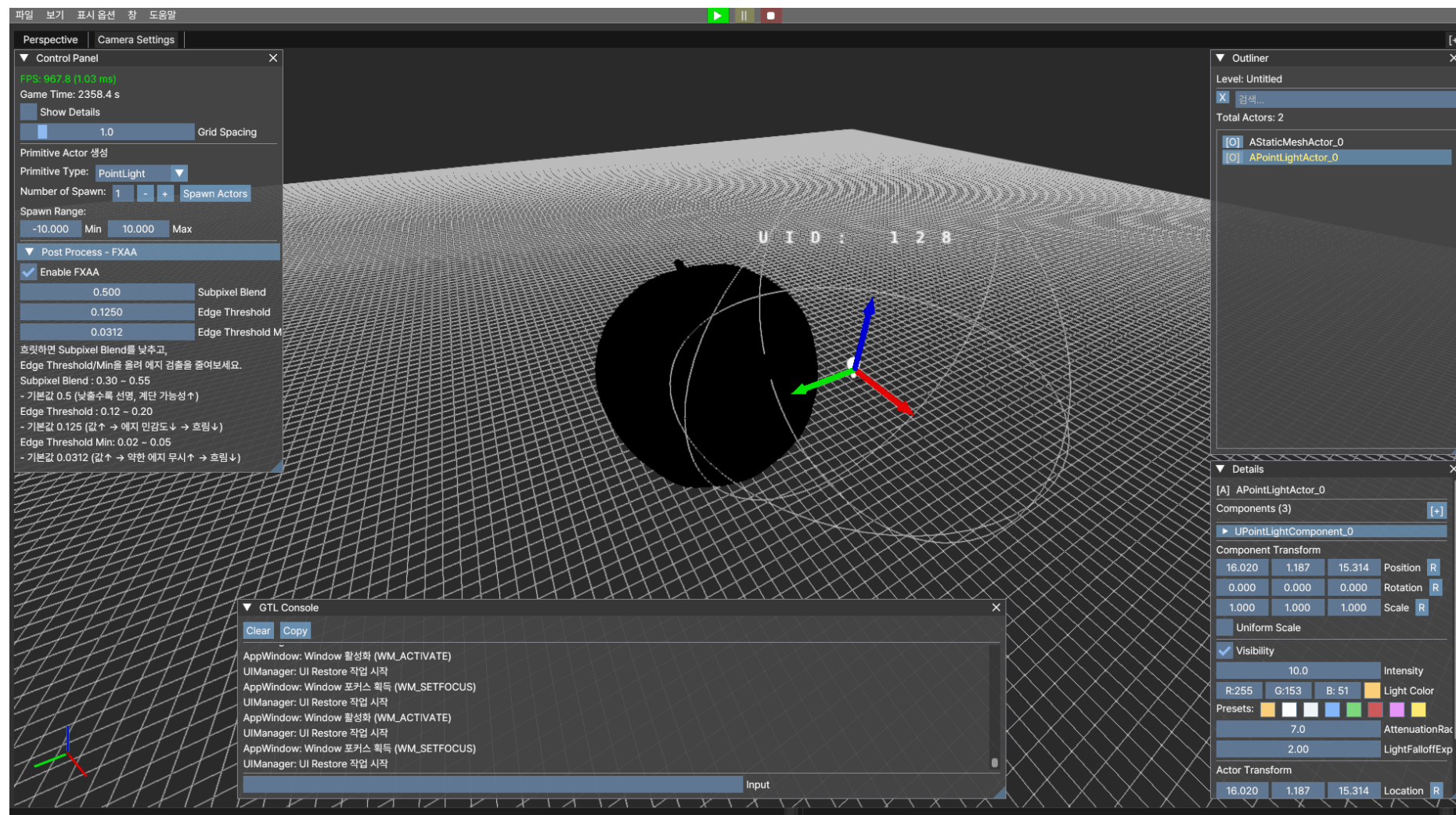
Normal Mapping

04

Tile-based Light Culling

Lights

Point

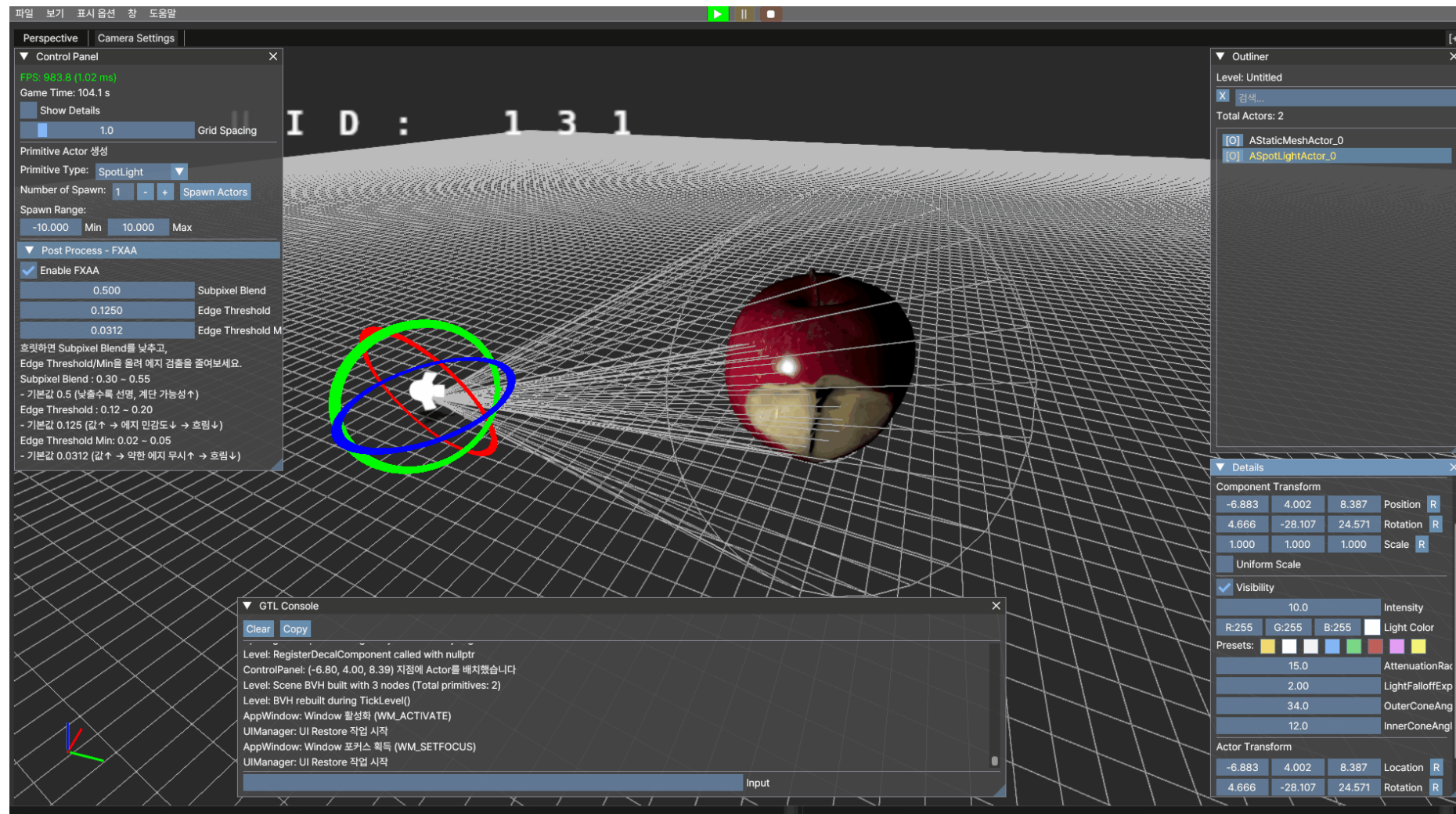


$$A(d) = \left(1 - \text{saturnate} \left(\frac{d}{R} \right) \right)^f$$

- 반경(R) , 감쇠지수 (Falloff)기반 거리 감쇠
- 반경에 가까워질수록 감쇠가 작아져서 조명이 부드럽게 감소

Lights

Spot

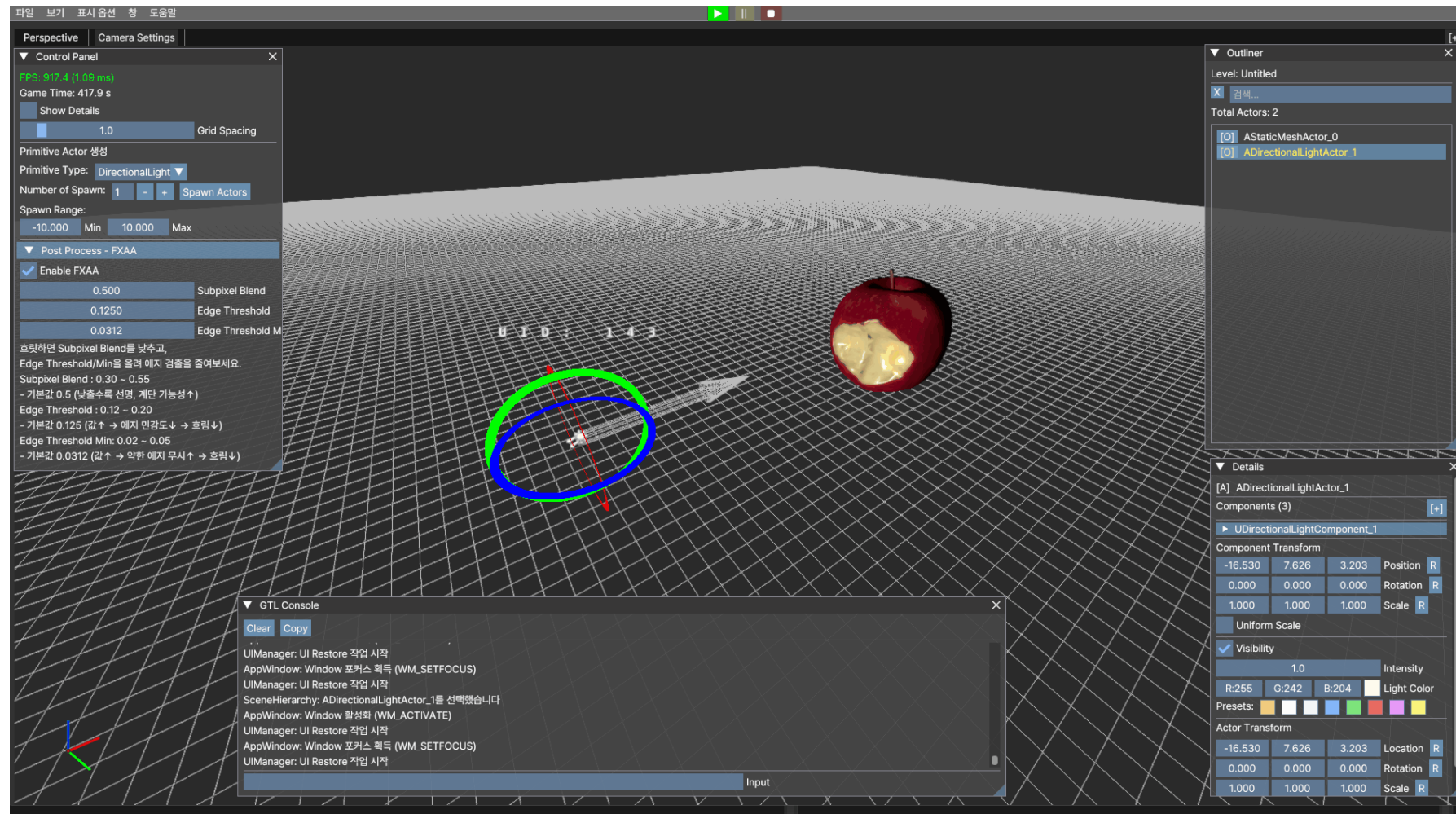


- Spot Cone : Inner / Outer 각도로 부드럽게 감쇠
- Distance Falloff : 거리 기반 밝기 감소
- Total Attenuation = 거리 감쇠 × 각도 감쇠

$$A_{spot} = \text{saturate} \left(\frac{\cos \theta - \cos(\text{Outer})}{\cos(\text{Inner}) - \cos(\text{Outer})} \right)^2$$

Lights

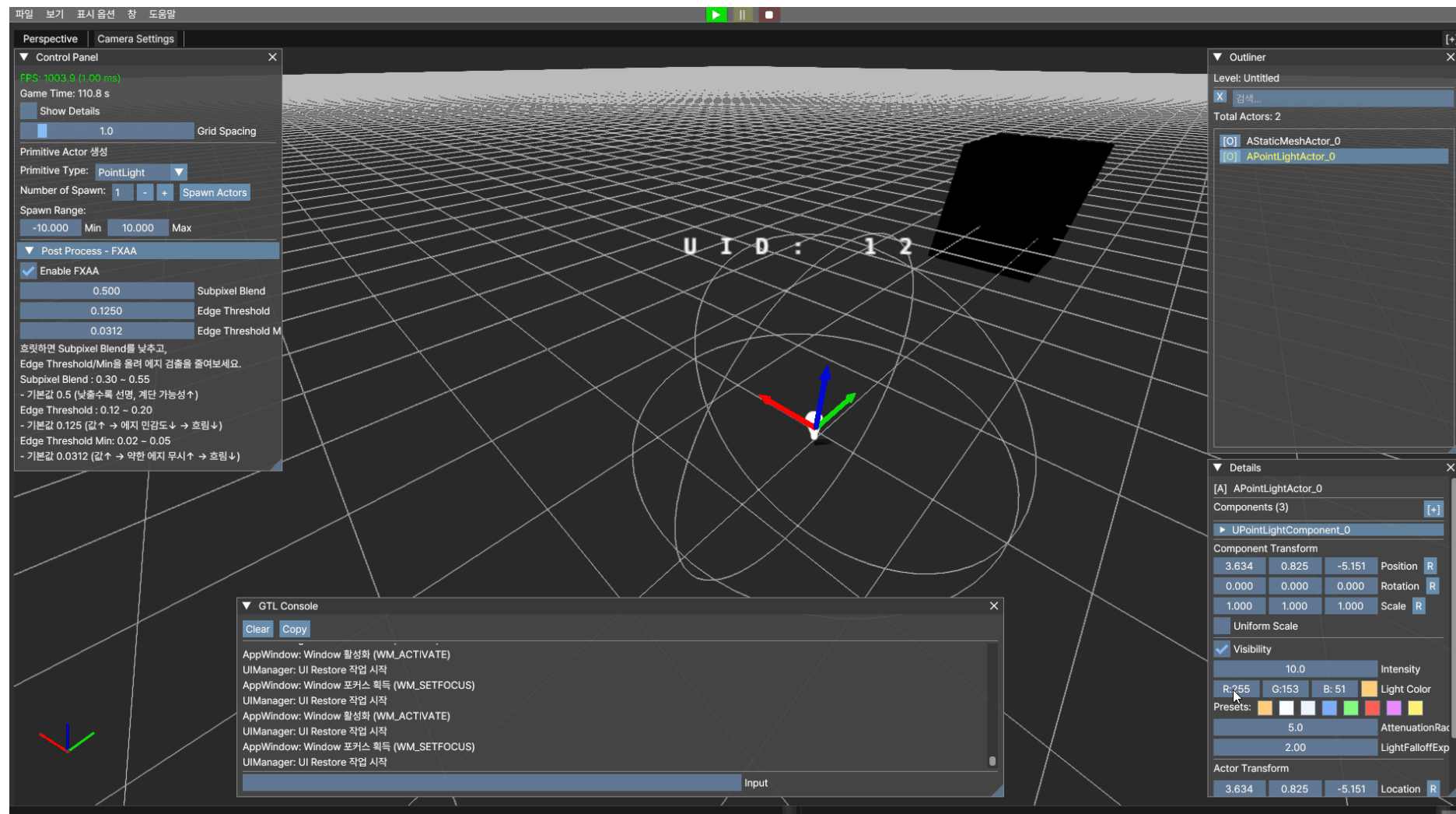
Directional



- 방향만 존재, 위치 없음 → 거리 감쇠 (Attenuation) 없음
- 씬 전역에서 동일한 세기의 빛

Shaders

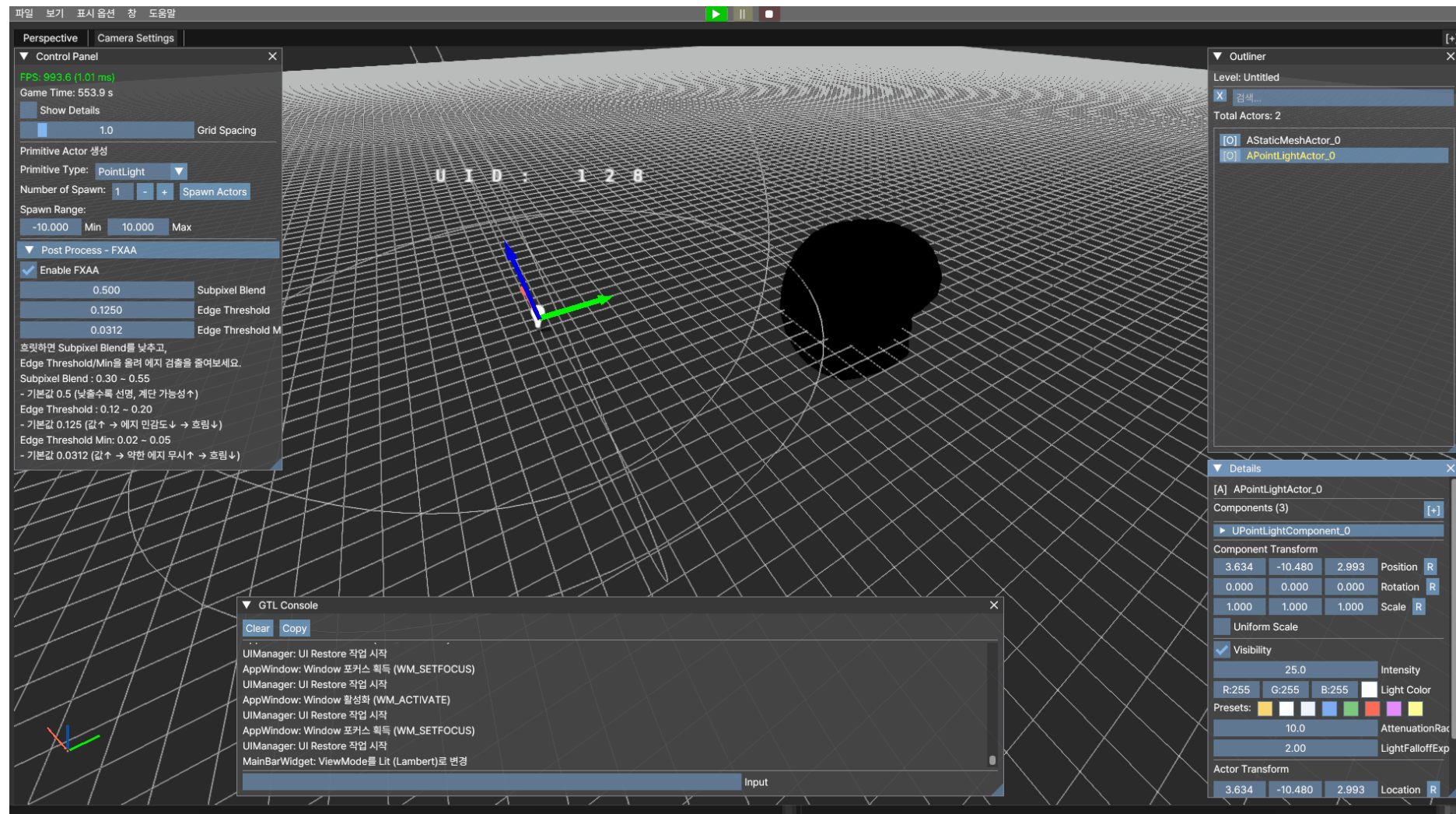
Gouraud



- 정점 단위 라이팅 → 픽셀로 보간
- VS에서 Ambient+Lambert 합산해 VertexColor 출력

Shaders

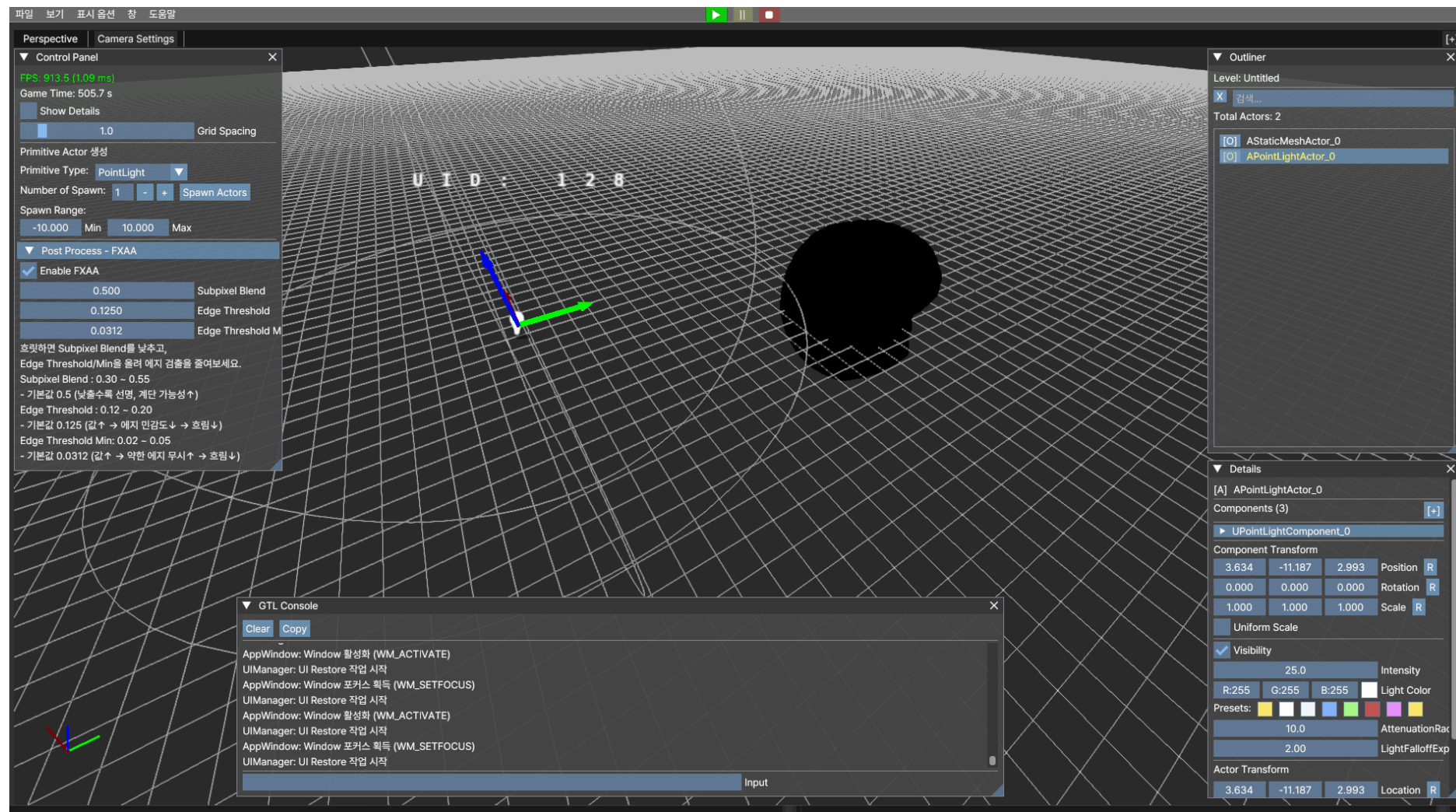
Lambert



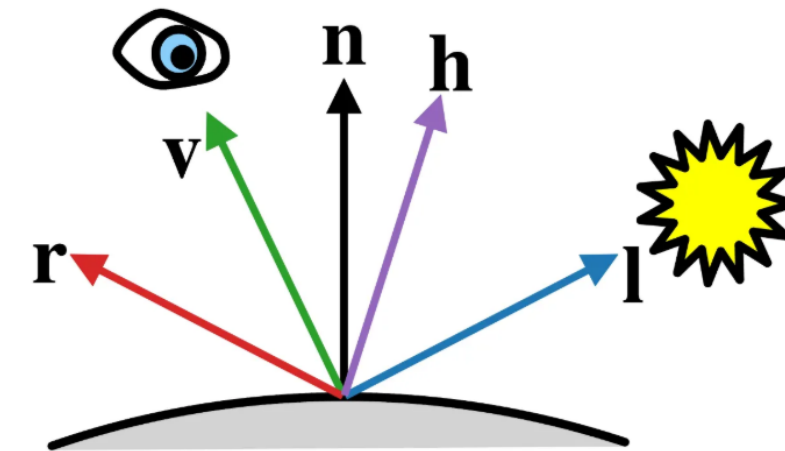
- PS에서 Ambient + Diffuse(난반사)를 계산
- 법선·광원 내적($N \cdot L$)로 밝기 결정, Specular 없음

Shaders

Blinn-Phong

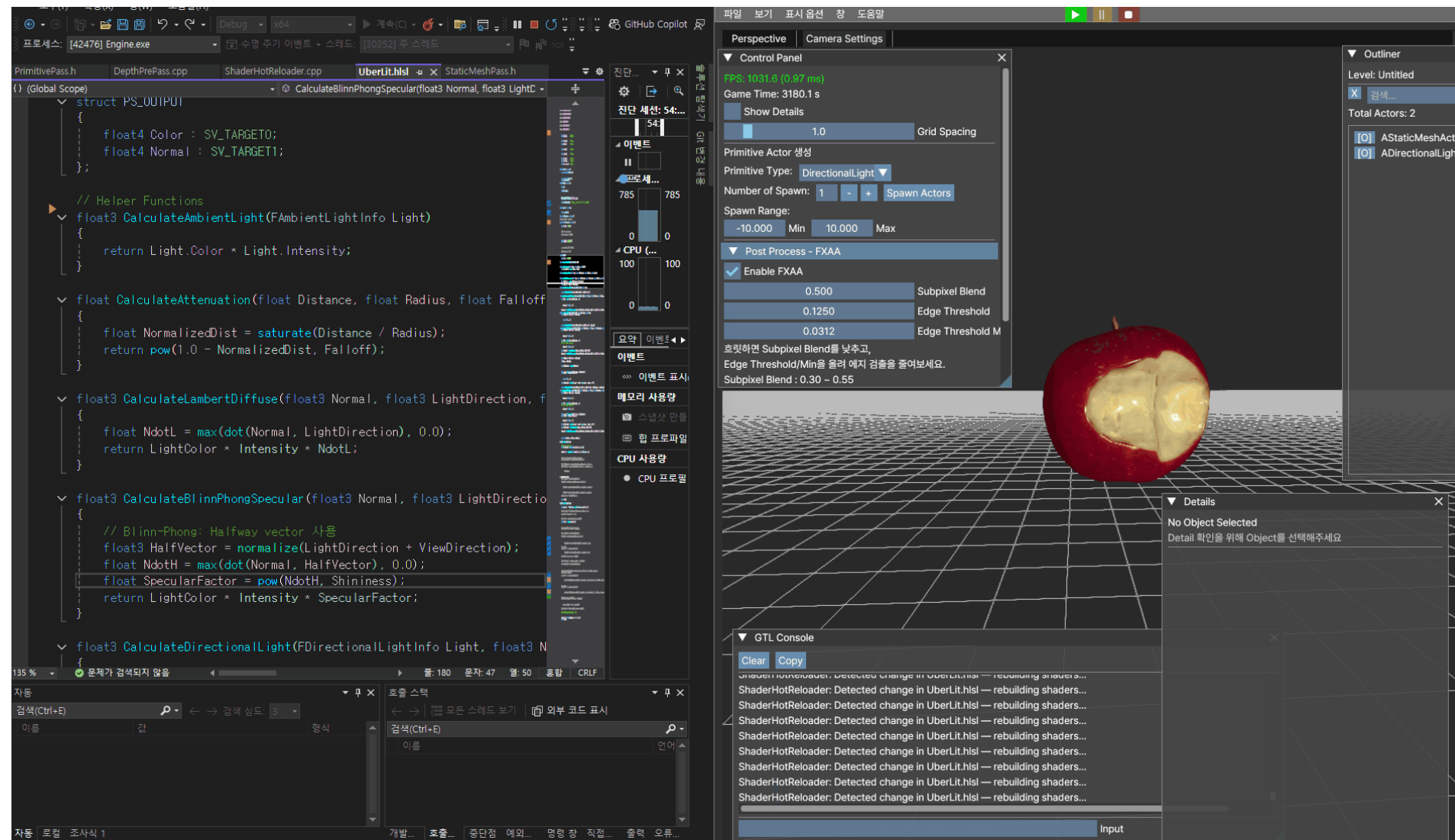


- PS에서 Ambient + Diffuse + Specular 계산
- 빛 방향과 시선벡터의 중간벡터, Half Vector로 정반사 표현



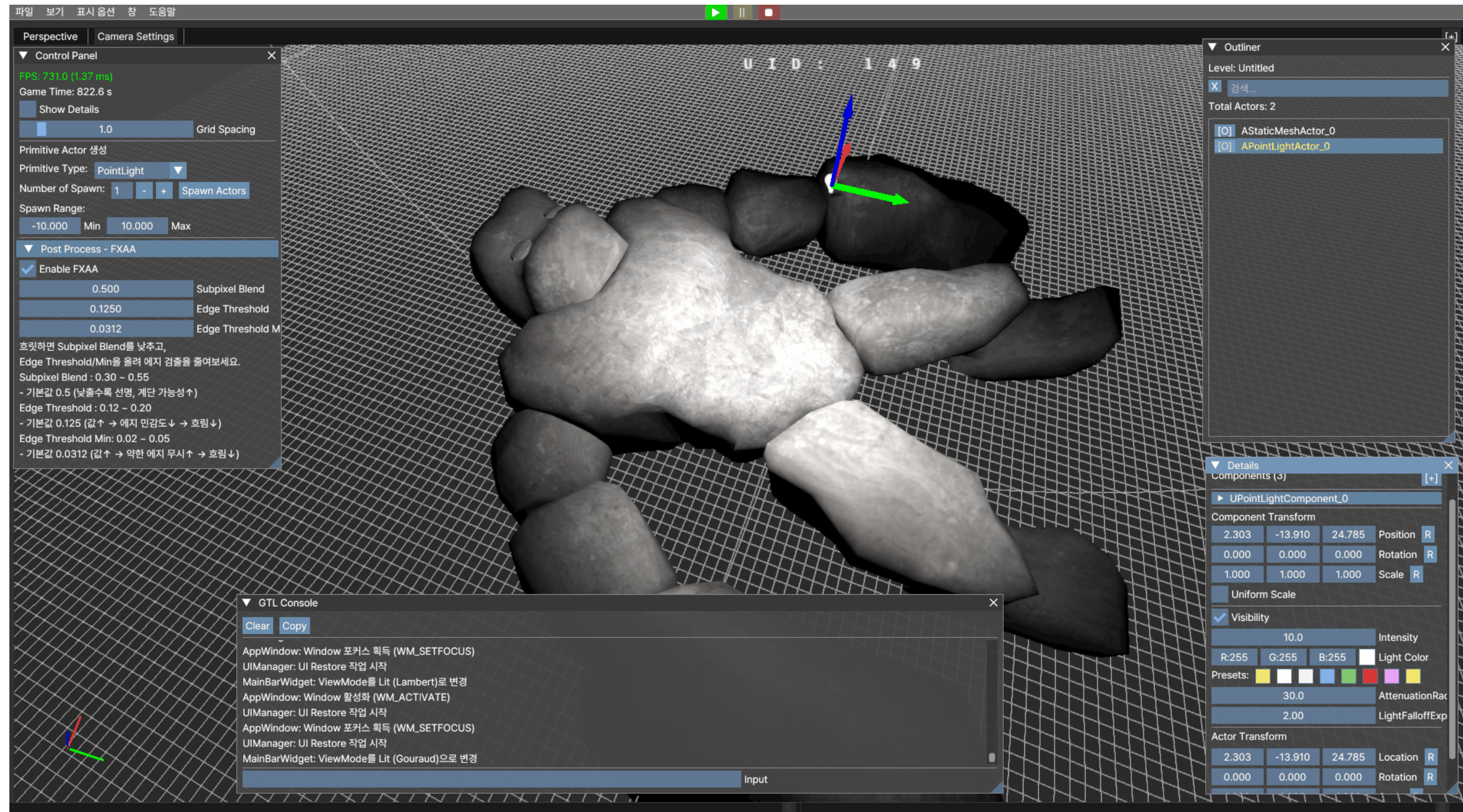
Shaders

Hot Reload

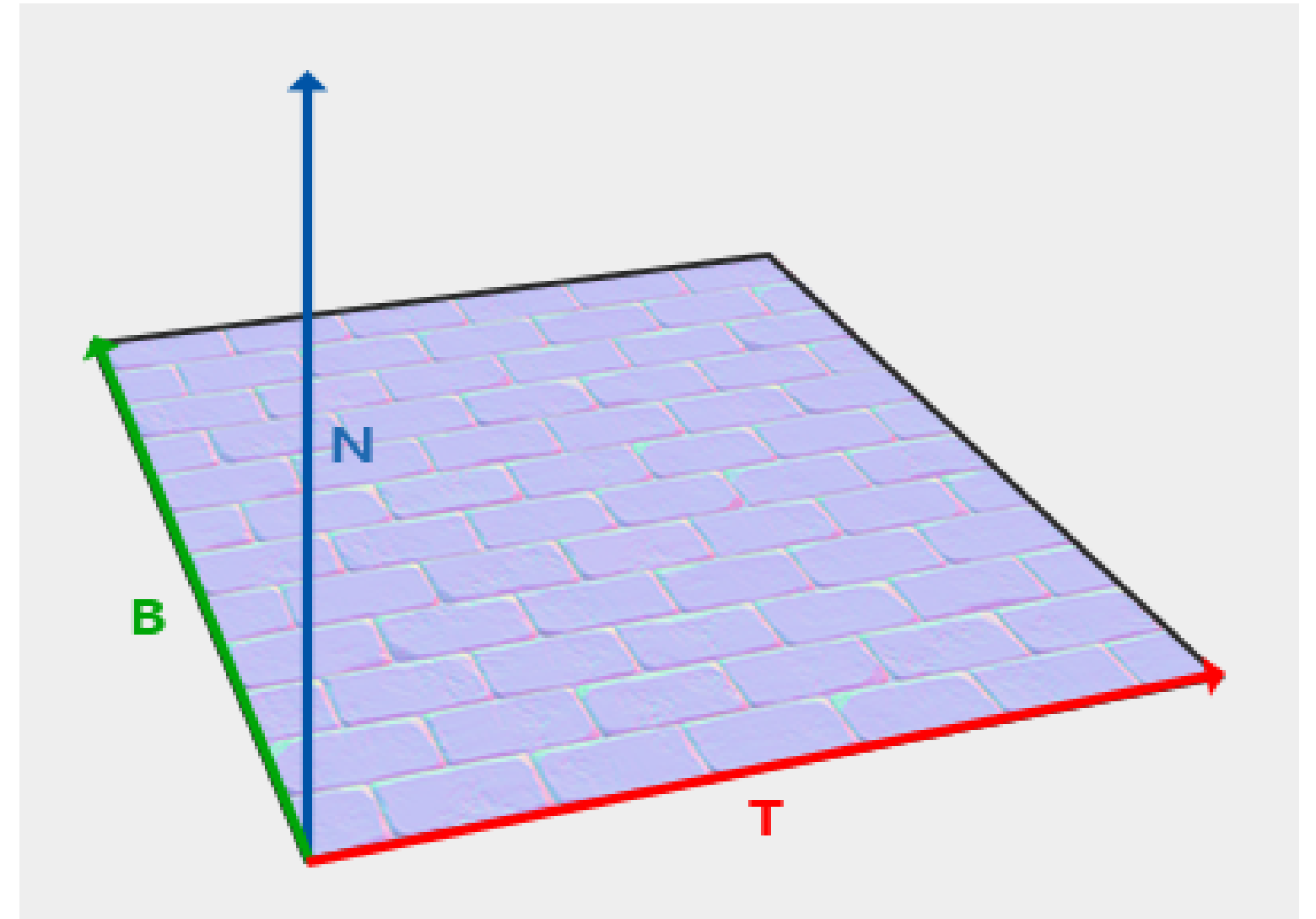
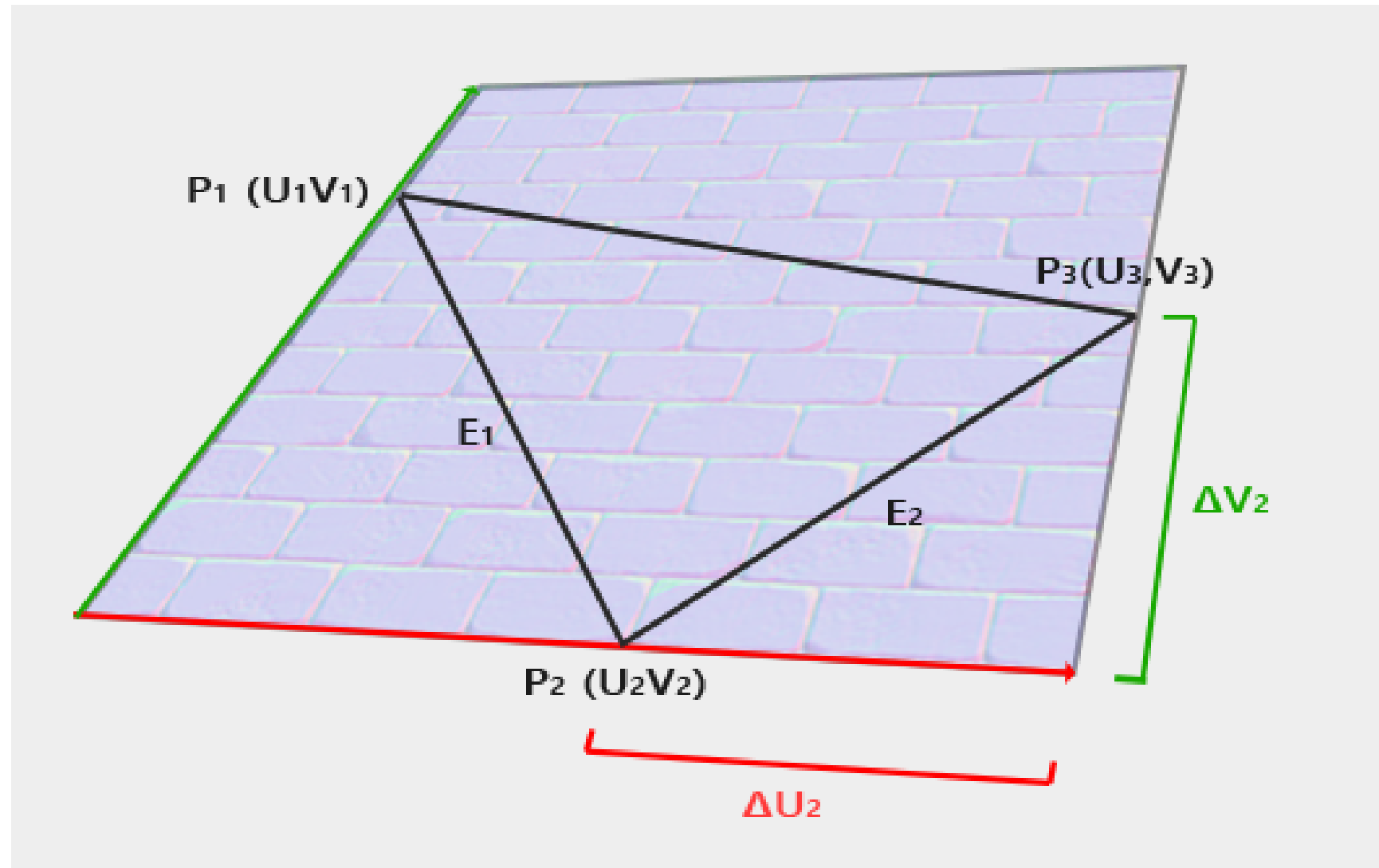


- 코드나 셰이더 수정 시 실시간 갱신
- Uber shader 수정 시 엔진 재시작 없이 즉시 테스트 & 디버깅 가능

Nomal Mapping



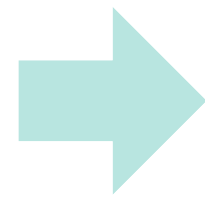
Normal Mapping



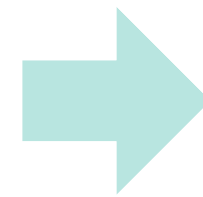
Nomal Mapping

$$\begin{bmatrix} T_x & T_y & T_z \\ B_x & B_y & B_z \end{bmatrix} = \frac{1}{\Delta U_1 \Delta V_2 - \Delta U_2 \Delta V_1} \begin{bmatrix} \Delta V_2 & -\Delta V_1 \\ -\Delta U_2 & \Delta U_1 \end{bmatrix} \begin{bmatrix} E_{1x} & E_{1y} & E_{1z} \\ E_{2x} & E_{2y} & E_{2z} \end{bmatrix}$$

폴리곤 각 정점의
UV 편차 + 법선 벡터

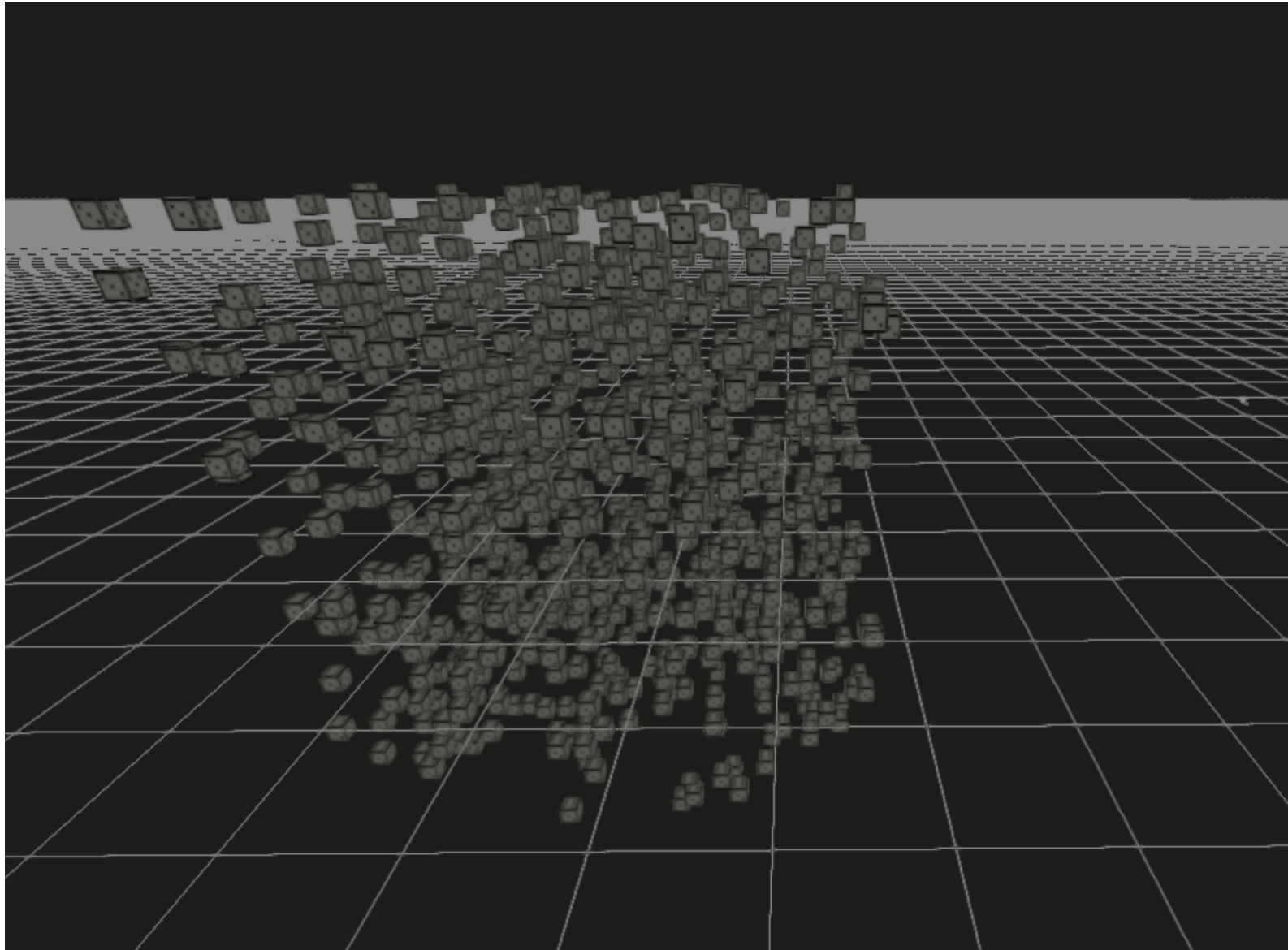


탄젠트 공간의
TBN 축



Normal Map 파일
의 법선을 World 공
간으로 변환

Tile-based Light Culling



- 스크린을 $32 * 32$ 사이즈의 타일로 분할
→ 각 타일에 대한 z-range 체크
- Depth 범위에 라이트가 들어오는지 체크
→ AABB & Sphere 충돌 순으로 판정

시연

<https://www.miricanvas.com/v2/design2/e4967b3c-0fc3-4448-8158-2c7ff7a26823>